Equilibrium and missing forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a missing force when acceleration is zero.

SUCCESS CRITERIA - I can...

1. Recognize rest and constant velocity as possible equilibrium.
2. Find missing forces from a zero net-force condition.
3. Explain why turning is not equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equilibrium” and “balanced” to explain your model or result. Sentence starters are optional.

equilibrium: zero net force and zero acceleration

balanced: adding to zero net force in this force context

tension: the pulling force through a stretched rope or cable

01 Warm-up: bring back an earlier idea

A 3 kg ball is moving upward at 5 m/s after leaving a hand. Ignore air resistance; use g=10 N/kg.
Draw and justify its free-body diagram.

Word help: justify = support your answer with a reason or evidence

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

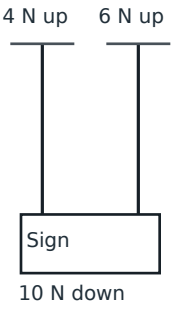
02 Worked example: explain the model

Reference: $\Sigma F_x = 0$; $\Sigma F_y = 0 \Rightarrow a = 0$.
Equilibrium allows rest or constant velocity. Resolve missing forces along each axis. Use g = 9.8 N/kg.

Worked reasoning: A sign has weight 10 N down and two vertical upward pulls of 4 and 6 N. The vector sum is zero, so equilibrium is possible.

Your turn: Draw and label all three forces.
Identify the equation that checks vertical balance.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Find the balance

The box moves right at steady velocity. A rope pulls 20 N right.
Find friction and net force.

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Steady rightward velocity

Marzano check - circle: ① ② ③

04 A still lamp

A lamp weighs 30 N and hangs from one vertical rope without accelerating. Add both forces to the diagram and find the rope tension.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label force arrows.

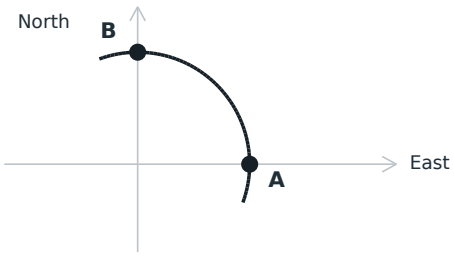
Stationary lamp: weight 30 N.

Marzano check - circle: ① ② ③

05 Parked or cruising?

Compare a parked car, a car moving straight at steady speed, and a car turning at steady speed. Which can have balanced forces? On the curve, add velocity arrows at A and B to explain the turning case.

Sentence starter: Both ... . They differ in ... because ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

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Continue down each column.

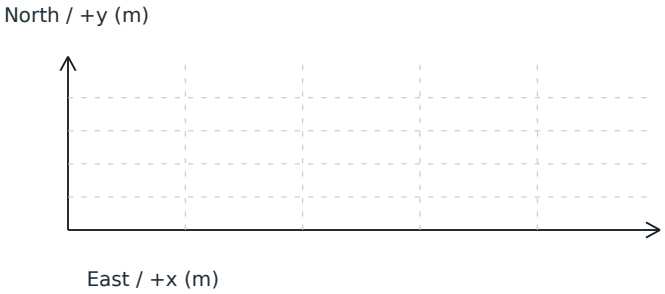
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A goes 7 m east and 5 m north. B then goes 4 m west and 9 m south.
Draw the route and find the endpoint relative to the start.
Draw the component vectors and resultant; label lengths.

Word help: component = the part of a vector along one chosen axis

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

07 Exit check

A box at rest has a 12 N force right and a 5 N force left.
Add the missing horizontal force needed for equilibrium.
Write an equation that checks the balance.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .



A third horizontal force is missing.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Return to the worked example in problem 2.
Find three different pairs of nonnegative upward pulls that balance the same 10 N sign.
Explain what stays constant.

Sentence starter: Both … . They differ in … because … .

Case	Upward pull A (N)	Upward pull B (N)
1	____	____
2	____	____
3	____	____

Marzano check - circle: ① ② ③